

Adjoint-Faithful Fourier Neural Operator for a Baroclinic Double Gyre

A minimum-data summer internship project plan

Core idea: first establish a frozen, adapted Bire-architecture baseline on the same MITGCM trajectories used by AF-FNO. Then improve the operator with modern forward losses and a small set of forward perturbation-response pairs. Generate true MITGCM adjoints only after the emulators are frozen, and use them as a blind test.

Revised technical proposal — 23 July 2026

Ten-week scope; official MITGCM baroclinic-gyre tutorial; NeuralOperator 2.0 implementation

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1 Recommendation in one page

Recommended path. Build an **adjoint-faithful FNO (AF-FNO)** on the official MITGCM baroclinic double-gyre tutorial at 1° . Before changing the architecture or loss, train a frozen **A0 adapted Bire FNO** on exactly the same trajectories. This tests transfer of the paper’s qualitative trends—useful short forecasts, stable rollouts, and sensitivity to wind amplitude—without claiming a bitwise reproduction of its undocumented 0.25° dataset. Then introduce the modern A–D ladder: A and B are controlled baselines, C is selected for forward fidelity, and D adds forward-only response supervision. Do not put a single MITGCM adjoint in training or model selection.

The minimum credible programme is:

- **Three long production regimes:** low, control, and high wind amplitude.
- **Two short held-out wind runs:** the intermediate amplitudes used by Bire et al.; inference-only and not 20 years long.
- **One frozen architecture-transfer baseline (A0):** the recovered Bire FNO core, adapted only where the 62×62 , 46-state benchmark forces a change. It is trained and evaluated before the modern A–D models.
- **224 ten-day perturbed restarts:** 128 training, 32 validation, and 64 inference. They provide local tangent information from forward MITGCM only and total about 6.1 model-years.
- **Twelve true-adjoint runs:** three smooth objectives, two horizons, and two unseen initial states, opened only after the model and criteria are frozen.

This separates four questions: Does the Bire architecture transfer to the documented tutorial data? Does modernization improve the forward emulator? Did response supervision improve the local input–output map? Does automatic differentiation through it yield a useful adjoint? The defensible claim is deliberately limited: *adjoint fidelity over specified horizons, objectives, states, and a tested control subspace.*

No-adjoint-training contract. MITGCM adjoints are excluded from training, hyperparameter tuning, early stopping, loss selection, perturbation-amplitude selection, and architecture selection. The test is run only after checksums of the frozen model and evaluation script are recorded.

2 Research question and hypotheses

Let the MITGCM time- Δt map be

$$x_{n+1} = \mathcal{M}_{\Delta t}(x_n, u_n),$$

where x is the state and u contains forcing and controls. The FNO $\mathcal{F}_\theta$ approximates the same map. For a scalar objective $J(x_N)$,

$$\lambda_n^{\text{FNO}} = \left(\frac{\partial \mathcal{F}_\theta}{\partial x_n} \right)^T \lambda_{n+1}^{\text{FNO}}, \quad \lambda_N = \nabla_{x_N} J.$$

Primary question. Can local derivative information obtained exclusively from short forward perturbation experiments make these gradients agree with MITGCM adjoints more closely than a state-accurate forward FNO?

Hypotheses: (1) the adapted A0 baseline reproduces the qualitative Bire forecast and wind-response trends on the coarser tutorial; (2) a forward-selected modern FNO improves forecast, rollout, boundary, and resolved-scale behavior relative to A0; (3) state accuracy alone does not ensure adjoint accuracy; and (4) adding a local response loss to a forward-credible model improves gradient direction without adjoint labels or material loss of forward skill.

3 Why local perturbation–response information is the right approach

3.1 A small state error can hide a wrong derivative

Suppose $\mathcal{M}(x) = x$, but an emulator learns

$$\mathcal{F}(x) = x + \delta \sin(kx), \quad \mathcal{F}'(x) = 1 + \delta k \cos(kx).$$

The state error never exceeds δ , yet the derivative error can be large when k is large. A value-only loss controls the height of a learned curve at sampled points; it does not adequately control its slope. An adjoint is built from those slopes.

Intuitive ocean example

Two emulators predict sea-surface height ten days ahead to within one centimetre. In one, a small extra wind-stress patch strengthens the western boundary current; in the other, it weakens it. Their forecast scores can be almost identical, but a gradient-based optimization steps in opposite directions. One perturbed forward run reveals the physical response sign without computing an adjoint.

3.2 Forward responses teach Jacobian actions

At an anchor $z = (x, u)$ and normalized direction v , run the model from z and $z + \varepsilon v$:

$$r_M(z; v, \varepsilon) = \frac{\mathcal{M}_{\Delta t}(z + \varepsilon v) - \mathcal{M}_{\Delta t}(z)}{\varepsilon} \approx D\mathcal{M}_{\Delta t}(z)v.$$

Match it with the emulator secant

$$r_\theta(z; v, \varepsilon) = \frac{\mathcal{F}_\theta(z + \varepsilon v) - \mathcal{F}_\theta(z)}{\varepsilon}.$$

This constrains a Jacobian–vector product (JVP) using only forward simulations and ordinary first-order training backpropagation. The link to adjoints is duality:

$$\underbrace{v^T D\mathcal{M}(z)^T q}_{\text{gradient projected on } v} = \underbrace{q^T D\mathcal{M}(z)v}_{\text{objective response along } v}.$$

Thus, matching $D\mathcal{M}v$ for relevant directions also matches projections of adjoints for many later objectives q . The full Jacobian remains too large to identify; the claim must stay within the sampled smooth wind and state subspace.

4 Benchmark choice: trend, not hidden dataset

The public `oceanfourcast` repository is a conceptual starting point, not an exact reproduction package. Its training defaults differ from the paper, its README points to external data, and its loader predicts ten selected channels including diagnostic pressure and streamfunction. Reimplement the method in the maintained `NeuralOperator` library rather than preserve old dependencies [4, 5].

Table 1: Published setup, official tutorial, and proposed benchmark.

	Bire et al.	Official tutorial	Proposed
Grid	0.25° , 248×248	1° , 60×60 ocean cells plus land border	Tutorial first; fine grid only as stretch
Vertical	15 levels, about 2 km	15 levels, 1.8 km	Tutorial levels
Time step	300 s	1200 s	Tutorial value
τ_0 (N m^{-2})	0.075, 0.0875, 0.100, 0.1125, 0.125	Default 0.100	Same five amplitudes
Training winds	Low, control, high	Not applicable	Low, control, high
Held-out winds	Two intermediates	Not applicable	Same two; short runs
Goal	Forward emulation	Numerical tutorial	Forward and local-adjoint fidelity

This removes uncertainty about an unavailable initialization and pipeline, reduces the horizontal cell count by about 16, and retains the relevant baroclinic physics. The tutorial notes that the deep ocean is still adjusting after 100 years; stationarity must therefore be measured for the variables and horizons used [3].

4.1 A0: a time-boxed Bire-architecture transfer baseline

Do not make the incomplete public repository a prerequisite for the project. Audit the paper, repository, and recovered implementation once; record unresolved discrepancies; then freeze A0 and label it *adapted* rather than exact. Implement it through the public `NeuralOperator 2.0 SpectralConv` interface while retaining the recovered paper structure: channel-diagonal Fourier multipliers, explicit pointwise channel mixing, three Fourier blocks, width 128, alternating sine/cosine position fields, and paper-style one-step state loss and optimizer schedule.

All models must use the same S0–S2 source states, chronological splits, masks, and training-only normalization. Only adaptations forced by the documented benchmark are allowed in A0:

Run a 20–100-sample overfit test, one ten-day development training, and the locked low/control/high evaluation. Do not tune A0 after opening held-out I1/I2 results. If the recovered baseline cannot learn the small sample after the declared learning-rate fallbacks, report that result and proceed; repairing historical code must not delay the AF–FNO experiment. Separate five- and 30-day A0 models are optional follow-ups only when needed to compare a specific Bire lead-time panel.

5 Minimum MITgcm simulation programme

Table 2: Declared adaptations from the Bire configuration to A0.

Element	Bire / recovered setting	A0 on the tutorial
Horizontal grid	248×248 at 0.25°	62×62 including the land rim
Dynamic state	Ten selected prognostic/diagnostic outputs	All 46 Markov-state channels; wind is the additional input
Fourier modes	[64, 64]	[16, 16], preserving the approximate mode-to-grid ratio
Coarse pretraining	Stride 8, yielding 31×31	Not used in the frozen A0; this is a recorded departure, not an inferred reproduction
Forecast target	Normalized next state	Retain for A0; Models A–D predict a normalized residual
Software	Historical oceanfourcast environment	Recovered architecture on pinned NeuralOperator 2.0

Table 3: Minimum trajectories. Durations are planning budgets; diagnostics decide whether equilibration is adequate.

ID	τ_0	Duration	Initialization	Use	Output
S0	0.100	100 y spin-up; final 10 y production	Tutorial initial state	Control data; branch checkpoint	Annual restart; daily production
S1	0.075	5 y adjust + 10 y production	S0 checkpoint	Low-wind data	Daily production
S2	0.125	5 y adjust + 10 y production	S0 checkpoint	High-wind data	Daily production
I1	0.0875	1 y	Unseen S0-derived state	Interpolation inference only	Daily
I2	0.1125	1 y	Different unseen state	Interpolation inference only	Daily
Total		142 model-years	Three long production regimes; two short inference regimes		

5.1 Five trajectory jobs; three long production regimes

After year 50, inspect five-year trends of heat content, surface kinetic energy, basin-integrated transport, and level-mean temperature. Stop at year 100 only if the upper-ocean quantities are statistically stationary; otherwise extend the run or narrow the first claim. The 1° grid and four-times-larger time step should be much cheaper per model-year than Bire’s setup, but wall-clock/model-year must be benchmarked.

5.2 Chronological data splits

Use years 1–7 for training, year 8 for validation, and years 9–10 for inference within each ten-year S0–S2 production segment. Insert 90-day gaps at boundaries.

Table 4: Data roles. Pair counts are approximate for daily output and a ten-day target.

Dataset	Source	Approx. size	Permitted use
Trajectory training	S0–S2 years 1–7	7,500 pairs	Fit weights
Trajectory validation	S0–S2 year 8	1,000 pairs	Tune and early-stop
Known-wind inference	S0–S2 years 9–10	2,100 pairs	Final scores only
Unseen-wind inference	I1 and I2	700 pairs	Final interpolation only
Response training	16 anchors $\times$ 8 directions	128 endpoints	Fit response loss
Response validation	4 anchors $\times$ 8 directions	32 endpoints	Tune loss and ε
Response inference	4 anchors $\times$ 8 directions $\times$ 2 signs	64 endpoints	Blind central differences
True adjoints	3 costs $\times$ 2 horizons $\times$ 2 states	12 integrations	Blind comparison only

The 224 perturbed restarts add 2,240 model-days, about 6.1 model-years. Save only initial/final states, configuration hash, and perturbation definition. A 46-channel 62×62 float32 snapshot is about 0.7 MB; 30 daily production years are of order 8 GB before compression.

5.3 Eight directions per response anchor

Table 5: Minimum smooth perturbation basis.

Count	Direction family	Purpose
1	Prescribed cosine-wind amplitude	Directly tests the Bire control axis
1	Smooth meridional wind phase/shape mode	Separates amplitude from pattern learning
2	Smooth localized wind envelopes	Probe western/southern and northern/interior control
4	Balanced, low-wavenumber state directions	Training-only EOFs or filtered anomalies; rotate among anchors

Pilot standardized amplitudes 10^{-3} , 3×10^{-3} , and 10^{-2} . Select the largest for which a central-difference linearity check is within 5–10%, then freeze the rule by direction family. Use randomly signed one-sided perturbations for training and central differences for the blind response test.

6 State, forcing, and preprocessing

6.1 Predict a Markov state, not independent diagnostics

Place the prognostic variables on a documented common tracer grid and stack vertical levels as channels:

$$x = [U_{1:15}, V_{1:15}, \Theta_{1:15}, \eta], \quad C_{\text{out}} = 46.$$

Vertical velocity, pressure, and streamfunction are not independent prognostic channels. Streamfunction is diagnosed from all 15 U levels. For the present z-coordinate, linear-EOS, constant-salinity configuration, pressure is reconstructed as the exact MITGCM diagnostic PHHYD, i.e. hydrostatic pressure anomaly divided by ρ_0 , in $\text{m}^2 \text{s}^{-2}$:

$$\frac{\rho'_k}{\rho_0} = -\alpha_T(\Theta_k - T_{\text{ref},k}), \quad \Phi'_k = g\eta + \mathcal{H}_k\left(\frac{\rho'}{\rho_0}\right).$$

Here $\mathcal{H}$ is the fixed finite-difference vertical integration implemented in `calc_phi_hyd.F`; the $g\eta$ term matches `diags_phi_hyd.F`. Report Φ' at Bire's zero-based surface, mid-depth, and bottom indices 0, 7, 14. Physical pressure anomaly in Pa is $\rho_0\Phi'$. This map is linear, differentiable, and has a fixed transpose $\mathcal{H}^T$ for later adjoint objectives. It therefore adds evaluation views but no new learned information or output channels. If salinity later becomes active and nonconstant, add its 15 levels and update the equation of state rather than silently omitting it.

The implementation was checked against the archived S0 MITGCM final-state PH dump at iteration 2,851,200 over all 15 levels and 3,600 wet cells. Maximum absolute error was $3.89 \times 10^{-4} \text{ m}^2 \text{s}^{-2}$ and RMSE was $9.75 \times 10^{-6} \text{ m}^2 \text{s}^{-2}$, passing the predeclared 10^{-3} maximum-error tolerance. The reproducible report is `outputs/af_fno/pressure_validation_v1/pressure_validation.json`.

Inputs comprise the 46 dynamic channels, wind stress, and normalized longitude, latitude, wet mask, and distance-to-wall fields. Record every C-grid interpolation operator. Compare gradients only after applying the exact transpose of those operators so the FNO and MITGCM sensitivities share a native control basis.

Compute channel means and scales from trajectory training only, mask land, and store the affine transform with the model. Gradient comparisons must undo normalization:

$$\tilde{x}_i = (x_i - \mu_i)/\sigma_i \implies \frac{\partial J}{\partial x_i} = \frac{1}{\sigma_i} \frac{\partial J}{\partial \tilde{x}_i}.$$

Models A–D predict the normalized ten-day residual $\Delta\tilde{x}_n = \tilde{x}_{n+1} - \tilde{x}_n$. This gives the network an explicit identity path and focuses capacity on change. A0 alone retains the recovered paper-style normalized next-state target; this intentional difference belongs to the architecture-transfer baseline and must be stated in every A0 comparison.

7 Modern adjoint-faithful FNO

7.1 Starting implementation

After freezing A0, use the maintained `neuraloperator` package and pin its release and commit. Begin the common A–D ladder with a modest dense FNO:

Select mode count from the resolved power spectrum: too few modes oversmooth the western boundary current; too many learn aliased fluctuations. Domain padding or Fourier continuation is relevant because the basin is not periodic. These choices follow the NeuralOperator 2.0 practical guide rather than the outdated oceanfourcast environment [2].

Table 6: Starting AF-FNO configuration. Validation uses only trajectories and forward responses.

Choice	Starting value	Reason / limited search
Forecast interval	10 days	Economical response runs; useful scale in Bire et al.
Fourier modes	[16, 16]	Compare 12, 16, 24 after inspecting spectra; respect Nyquist
Hidden channels	32	Increase to 64 only if a 20–100-sample overfit test fails
FNO layers	4	Current library default; compact enough for ablation
Local branch	<code>conv_bias_kernel=3</code>	Complements global mixing near boundaries
Boundary treatment	10% domain padding	Reduces periodic FFT wrap in a closed basin
Geometry	Grid embedding, mask, wall distance	Makes solid boundaries explicit
Channel MLP	Enabled	Mixes the vertical state channels locally
Precision	float32	Avoid mixed precision in the initial derivative study
Factorization	Dense FNO first	TFNO is a memory fallback, not an initial confounder

7.2 Baseline ladder and controlled losses

For a one-step pair and response direction, use

$$\mathcal{L} = \mathcal{L}_{\text{state}} + \lambda_{\text{roll}}\mathcal{L}_{\text{roll}} + \lambda_{\text{resp}}\mathcal{L}_{\text{resp}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{bnd}}\mathcal{L}_{\text{bnd}}, \quad (1)$$

$$\mathcal{L}_{\text{resp}} = \frac{\|W_r(r_\theta - r_M)\|_2^2}{\|W_r r_M\|_2^2 + \epsilon_0}. \quad (2)$$

$\mathcal{L}_{\text{state}}$ is masked relative L_2 ; $\mathcal{L}_{\text{roll}}$ is a three-step (30-day) unrolled loss; $\mathcal{L}_{\text{spec}}$ compares binned spectra; and $\mathcal{L}_{\text{bnd}}$ upweights wet cells near the western wall. An H^1 -style spatial loss may supplement the spectral term, but it is not a substitute for $\mathcal{L}_{\text{resp}}$: spatial output derivatives and derivatives of the input–output operator are different objects.

Table 7: Required architecture-transfer baseline and controlled model ladder.

Model	Training signal	Purpose	Expected result
A0	Adapted Bire architecture and paper-style one-step loss	Test architecture transfer before modernization	Qualitative Bire trend on the common data
A	Modern dense FNO + one-step residual state loss	Controlled modern baseline	Comparable state skill; derivatives may be poor
B	Model A + rollout + spectral/boundary losses	Isolate modern forward losses from response data	Better rollout and boundary state skill
C	Forward-selected dense FNO + balanced state/increment, rollout, tapered spectral, and boundary losses	Establish a forward-credible emulator before derivative training	Pass the complete forward gate
D	Frozen Model C design + local response loss	Test the AF-FNO hypothesis without asking derivative data to repair the forward model	Preserve C’s state skill; improve responses and adjoints

Models A and B remain frozen controls; B is not retroactively retuned. Model C is the only forward-selection stage, and its search is limited and recorded below. Model D copies C’s selected architecture,

preprocessing, and forward objective, then adds $\mathcal{L}_{\text{resp}}$. Increase λ_{resp} only on response validation data until response error improves without worsening C's validation state error by more than 5%. Lock all weights before viewing any true adjoint.

7.3 Model C: predeclared forward-selection protocol

The preliminary A/B results show that architecture plausibility is not enough: temperature and SSH can be worse than persistence even when the 46-channel aggregate loss improves. Following the practical FNO workflow in Duruisseaux et al. [2], Model C therefore separates representation, optimization, and data adequacy rather than changing all three at once.

Model C implementation contract opened 23 July 2026. The first implementation retains the 51 declared raw inputs, NeuralOperator grid lifting, dense float32 spectral weights, Channel MLP, four-layer width-32 starting point, 10% padding, and parallel 3×3 local branch. It replaces B's channel-aggregated objective with equal U/V/temperature/SSH group terms, scales increment error with training-only per-channel ten-day RMS, and compares spectra of increments on the exact 60×60 wet rectangle after a Hann taper. Initial forward/backward checks contain 630,262 parameters at width 32 and 2,443,542 at width 64. Width 64 is therefore a conditional capacity test, not the assumed default. Training-only C1a calibration subsequently froze increment/rollout/spectral/boundary coefficients 0.001/0.15/ 10^{-5} /0.065; these scales precede all validation and architecture selection.

C0 training-only audit, 23 July 2026

The sealed audit verified the dataset metadata hash and read only 7,530 training pairs. The per-channel normalized ten-day increment RMS spans a factor of 147, so channelwise increment scaling is necessary. At (16, 16), the global Fourier path contains 8.8%, 24.0%, 67.8%, and 85.6% of tapered U, V, temperature, and SSH increment energy; full-state fractions all exceed 99.6%. The local branch is therefore essential, and more modes alone cannot represent most velocity-increment energy. The (24, 16) candidate captures 12.5%/32.9% for U/V, compared with 9.0%/25.6% for (16, 24); it was added before validation was opened. Increment autocorrelation implies only about 278–398 effective samples across the three regimes, with S1 much more correlated than S0/S2. This supports regime-balanced learning curves, but not immediate replacement of version-1 data.

C1a loss-gradient calibration, 23 July 2026

Complex-safe CPU job 285116 completed in 4m39s using only the same 96 balanced training rollouts. After its 20-epoch core warmup, state/increment/rollout/spectral/boundary gradient norms were 1.965/830.5/6.484/40,763/7.411. The automatically clipped proposal was rejected: coefficients 0.01 would still make increment and spectral gradients dominate. Unclipped targets were 0.001183/0.1515/ 1.205×10^{-5} /0.06629; the rounded frozen coefficients above produce weighted gradient ratios 0.423/0.495/0.207/0.245 relative to state. The very small spectral coefficient does not remove spectral supervision because its raw gradient is about 20,700 times the state gradient. Pending GPU replica 284858 was cancelled with zero runtime. No candidate checkpoint or validation metric was produced.

C1b memorization result and C1c duration test, 23 July 2026

Training-only job 285192 completed both controlled 160-epoch attempts in 58m30s. At learning rate 0.001 it passed total, spectral, state-group, rollout, and boundary criteria, but temperature/SSH increment errors were 1.802/2.888 times persistence on the exact 96 records. Learning rate 0.0005 improved these to 1.285/2.032; U/V were already 0.254/0.442 times persistence. Therefore C1b is rejected specifically for slow-field increments despite a 97.7% total reduction, 99.1% spectral reduction, 0.0304 maximum state-group error, 0.0312 rollout, and 0.0397 boundary error.

The lower learning rate achieved its best total at the final epoch, so undertraining is tested before changing loss weights or capacity. C1c fixes the same data, loss, architecture, seed, and batch order, uses only learning rate 0.0005, and extends 160 to 320 epochs. Its corrected gate compares every increment group with persistence measured on the exact sampled records rather than assuming that the full-training RMS makes every finite subset exactly one. Rejection now preserves learning curves and a bitwise-reload-tested diagnostic checkpoint. No validation data are opened.

1. **Training-only diagnostics.** Compute tapered two-dimensional spectra of normalized states and ten-day increments by variable group, verify padding and mode indexing, run a stronger 96-sample memorization test, and log both raw and weighted loss terms and their gradient norms. Report U, V, temperature, and SSH separately; the 15/15/15/1 channel multiplicity must not decide the fit.
2. **Repair the objective before enlarging the network.** Start from B’s dense four-layer, width-32 architecture. Use an equal-weight mean of the four group-normalized state errors, add a group-balanced normalized-increment error, retain the three-step rollout, and compute spectral loss after wet-field extension and a fixed taper so the land discontinuity is not mistaken for unresolved physics. Normalize boundary and spectral terms by group and choose their weights from the training diagnostics, then freeze the weights for candidate comparison.
3. **Small representation search.** Compare mode pairs (12, 12), (16, 16), (16, 24), (24, 16), (24, 24) and widths 32 and 64 with four layers and 10% padding. Use successive halving on fixed training/validation blocks. For only the two best candidates, test padding 20% and six layers if the failure diagnostic specifically indicates wall leakage or insufficient capacity. Dense float32 spectral weights, lifting, projection, Channel MLP, mask, wall distance, and the local 3×3 branch remain mandatory; factorization is a memory fallback.
4. **Freeze by validation, then test once.** Train the selected configuration with three fixed seeds. Candidate ranking is lexicographic: first require every U/V/temperature/SSH ten-day validation RMSE ratio to beat persistence; next minimize the worst group ratio; then minimize the predeclared mean of 30-, 90-, and 180-day rollout, boundary, and resolved-spectrum scores. Inference years, I1/I2, response inference, and adjoints remain sealed until the configuration, seeds, code, and normalizer hashes are recorded. The complete 10–360-day package is run only after that freeze.

This search is deliberately not “try until a test plot looks good.” A candidate that reduces the aggregate validation loss but remains worse than persistence for any state group is not selected. Conversely, increasing Fourier modes is not automatic: the 62-point grid limits meaningful frequencies, and added modes are accepted only when training-only increment spectra and held validation metrics support them. For the real-valued NeuralOperator convolution, a declared $\mathbf{n_modes} = (m_y, m_x)$ retains m_y centered Fourier rows but only $m_x/2 + 1$ nonnegative real-FFT columns internally. Every spectral audit and report uses that exact convention rather than interpreting 16×16 as 256 independent complex coefficients.

Calibration backend rule. CUDA is an efficiency choice, not a scientific requirement for the C1a loss-gradient calibration. Its 20 epochs on 96 training rollouts completed on eight CPU cores in 4m39s.

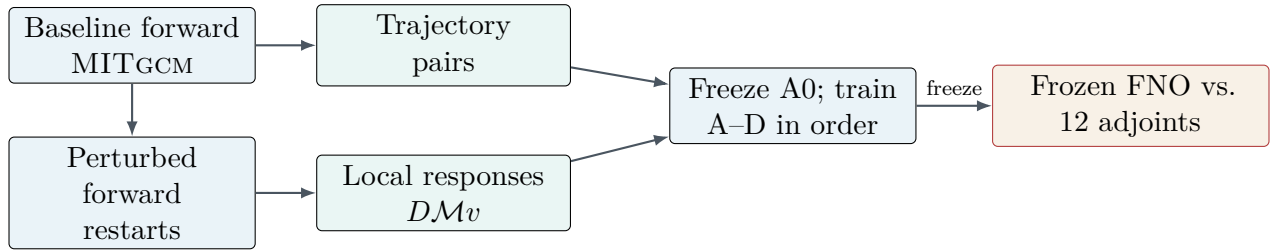
The CPU and GPU replicas must use the same data, records, seed, architecture, and loss code but write separate artifacts. If both complete, compare raw terms, gradient ordering, and proposed weighted contributions; modest floating-point differences are expected, whereas an order-of-magnitude disagreement blocks the loss freeze and requires diagnosis. Global gradient norms are accumulated in float64 for real gradients and complex128 for complex spectral-weight gradients even though model arithmetic remains float32/complex64; this prevents diagnostic overflow without discarding the imaginary component. Neither backend output is a candidate checkpoint or validation result.

7.4 Data-adequacy and expansion gate

The existing version-1 trajectories are sufficient to diagnose and select Model C: across S0–S2 they provide 7,530 overlapping ten-day training pairs, 780 validation pairs, and 1,860 sealed inference pairs. They are not 7,530 independent realizations, because adjacent daily pairs overlap strongly. More consecutive years would therefore increase the raw count much faster than the effective sample size.

Before generating replacement data, run learning curves on chronological 25, 50, 75, and 100% subsets, estimate integral autocorrelation times for the four state groups, compare training and validation error by group, and measure the three-seed spread. The evidence is considered *data-limited* only when a candidate can fit the training blocks, validation skill improves consistently with more chronology, and between-seed or between-regime generalization remains the dominant error. Poor training fit or insensitivity to added chronology points instead to optimization, capacity, or the objective and does not authorize more MITGCM years.

If this gate supports expansion, create a versioned `trajectories_v2`; never overwrite version 1 or its results. First target twice the *effective* independent coverage using additional branches from separated equilibrated checkpoints in every wind regime, rather than merely appending adjacent years. Consider three times the coverage only if the two-times learning curve remains materially positive. Preserve a sealed test segment and 90-day buffers. Any reported data-scale comparison retrains the relevant baselines on v2 or labels the changed-data comparison explicitly. Intermediate winds I1/I2 and perturbation/adjoint truth are still generated only after the forward choice is frozen.



The adjoint archive has no path into training or tuning.

Figure 1: The common trajectory pipeline feeds A0 and A–D; response data enter only D, and the adjoint evaluation remains sealed.

8 Ground-truth MITgcm adjoint suite

8.1 Minimum: twelve integrations

One adjoint integration returns sensitivities to the full initial state and active controls for one scalar objective. It is therefore more efficient to vary objective, horizon, and state than to run an adjoint for every perturbation direction.

Table 8: Blind matrix: 3 objectives $\times$ 2 horizons $\times$ 2 initial states = 12 adjoint integrations.

Smooth scalar objective	Reason	Horizons	States
Gaussian interior sea-surface-height average	Broad low-wavenumber response away from walls	10, 30 days	Two unseen dates
Smooth western-boundary sea-surface-height average	Energetic region where spectral/boundary errors matter	10, 30 days	Same dates
Area-weighted barotropic gyre-strength functional	Dynamical circulation measure; avoids nondifferentiable maxima	10, 30 days	Same dates

Select dates and masks before opening any adjoint result; place dates at least 90 days from split boundaries and one another. If mandatory tests succeed, an optional six-run extension adds 90-day horizons. Before accepting a reference adjoint, run a Taylor remainder or central directional finite-difference check at one test state per objective family. This verifies MITGCM but may not alter the emulator.

8.2 Metrics and an operational test

For gradients g_M and g_F , report

$$\text{cosine} = \frac{g_F^T g_M}{\|g_F\|_2 \|g_M\|_2}, \quad \text{angle} = \cos^{-1}(\text{cosine}), \quad \text{relative error} = \frac{\|g_F - g_M\|_2}{\|g_M\|_2}.$$

Convert both gradients to the same physical units and grid with explicit area/volume weights. Report full-state scores and projections onto the directions in Table 5. Direction is primary: optimization can tolerate imperfect magnitude but not a gradient pointing uphill. Also show signed maps, vertical profiles, wavenumber spectra, and sign agreement in the largest-sensitivity region.

Finally, use the emulator gradient to propose a small control update, choose the step from a predeclared short line search, and evaluate the objective with one forward MITGCM run. This tests usefulness without turning adjoints into labels.

9 Evaluation and predeclared success criteria

9.1 Bire et al. as a qualitative forward benchmark

The numerical errors in Bire et al. cannot be transferred to this experiment: their 0.25° , more turbulent double gyre has 248×248 points and different state variables, while the present tutorial has 62×62 points. We instead predeclare the following *trends* from their complete forward study as the benchmark [1]:

1. Their ten-day map gave the best overall compromise among 5-, 10-, and 30-day maps. It beat persistence and pointwise time-mean climatology at useful short leads; velocity lost deterministic skill first, near roughly three months, while pressure and surface temperature retained useful skill longer.
2. Spatial error grew first in the energetic western-boundary region. At day 60 the ten-day map retained the most credible large-scale and boundary structure; after deterministic phase information was lost, stable long integrations still retained the characteristic anomaly length scales.
3. The learned integrations remained bounded for thousands of model days. The five-day map developed excessive low-wind mesoscale variability, whereas the 30-day map aliased or missed propagating

variability whose period was close to the map interval. This makes spectra and longitude–time diagrams necessary alongside scalar RMSE.

4. Long-run maximum streamfunction increased approximately linearly with wind-stress amplitude, close to the expected Sverdrup trend for the ten-day map. The low- and high-intermediate wind cases were held out of training, so their ordering was a real interpolation test rather than a fit diagnostic.
5. Their emulator did *not* reliably steer from an initial condition from one regime after an abrupt change in imposed wind. Initial-condition dynamics dominated the new forcing. We record this negative result explicitly: matching the equilibrium wind ordering is necessary, but a response/adjoint model intended for control should also improve the forcing-switch response.

“Reproduce or improve Bire’s trends” below means recover this evidence pattern, not match their plotted values or claim an exact reproduction. A model may decorrelate—a chaotic forecast should—but it must beat both simple baselines before decorrelation, remain bounded, preserve resolved spatial scales, and express the correct forcing ordering. The complete adjoint campaign does not open until at least one modern forward model passes the gate below.

9.2 Frozen forward evaluation package

Evaluate A0 and Models A–D on identical chronological records and identical 15-member ensembles in each forcing regime. A0 is frozen before I1/I2 are opened and no model is tuned after viewing response or adjoint tests. Persistence and a regime-specific, training-only pointwise mean are the two primary baselines. The current configuration has no imposed seasonal cycle, so calling the latter a daily climatology would be misleading; a day-of-year climatology will be added only if seasonal forcing is added.

Every frozen model must produce the following versioned package. Evaluation contract version 2 adds the deterministic pressure postprocessor; version-1 packages remain historical pre-pressure evidence and are never overwritten. Machine-readable JSON stores definitions, scalar summaries, start indices, hashes, and gate outcomes; compressed NPZ stores the plotted arrays so a figure can be regenerated without rerunning the network.

1. **Deterministic skill:** physical-unit RMSE and pointwise-climatology ACC from 10 to 360 days for U, V, temperature, SSH, surface speed, SST, PHHYD at levels 0, 7, and 14, and barotropic streamfunction. Curves show ensemble mean and 10–90% range for the model, persistence, and climatology. Report the first lead at which each method loses ACC or baseline skill; never infer stability from the ten-day score alone.
2. **Spatial failures:** common-scale truth/forecast/error snapshots and spatial RMSE maps at 10, 60, 180, and 360 days for all seven Bire-facing diagnostics: surface speed, SST, SSH, the three pressure levels, and barotropic streamfunction. Report the first four wet cells from the western wall separately, plus full-domain vertical RMSE profiles, to expose boundary and deep-ocean drift.
3. **Stability and scale retention:** one-year trajectories of a surface kinetic-energy proxy, temperature, SSH, and maximum streamfunction; finite-value and land-mask checks; normalized mean/variance drift by depth; radially binned spectra for all seven Bire-facing diagnostics at 10, 90, 180, and 360 days; and a fixed-control-member SSH longitude–time diagram. Spectra are interpreted only over bins with non-negligible truth energy.
4. **Forcing physics:** low/control/high curves of maximum streamfunction, surface speed, SST, and SSH, with model and MITGCM slopes computed on the same ensemble. Once generated, I1/I2 are added as inference-only interpolation points. A separate forcing-switch experiment initializes every branch from the same control states and changes only the prescribed wind; it is clearly labeled as a

response diagnostic until matching switched-wind MITGCM trajectories exist.

5. **Audit trail:** checkpoint and dataset hashes, software/device metadata, exact ensemble starts, masks, normalizers, units, climatology definition, spectra, snapshots, maps, and all per-member values needed to reproduce uncertainty bands.

The forward gate is passed by a modern model only if all of the following hold:

- it improves on A0 and beats both persistence and training climatology at ten days for all seven Bire-facing diagnostics (speed, SST, SSH, surface/mid/bottom **PHIHYD**, and streamfunction), with useful ACC persisting beyond a single map step;
- its 360-day rollout is finite, respects the land mask, has no wet-cell normalized state magnitude above 20, and has no group-mean 360-day drift larger than two training standard deviations relative to the matching MITGCM ensemble;
- at 360 days its resolved-bin median spectral-energy ratio lies in $[0.25, 4]$ for each Bire-facing field; phase agreement is not required after decorrelation;
- the fitted low/control/high maximum-streamfunction slope has the same sign as MITGCM and magnitude within a factor of two. The same requirement is later applied with I1/I2 inserted, without refitting or retuning the emulator;
- the forcing-switch response has the physically expected sign by one year. Until switched-wind MITGCM truth is generated, this last item is reported as provisional and cannot by itself establish a pass.

Failure of a subcriterion is retained and plotted, not averaged away. In particular, a lower aggregate normalized loss cannot compensate for temperature or SSH being worse than persistence, a blow-up, or reversed wind sensitivity. If neither A nor B passes, the project proceeds only through the predeclared Model C forward-selection stage. The expensive perturbation/adjoint campaign remains closed unless C, or an explicitly versioned successor approved under the same validation-only rules, passes.

The three pressure fields are correlated derived views of temperature and SSH, not three additional independent trials. They are nevertheless required because Bire et al. reported them directly and because a model can have acceptable bulk temperature/SSH error while producing vertically accumulated hydrostatic errors that are unsuitable for pressure-based objectives. Gate failures are therefore interpreted field by field; pressure does not alter training or inflate the learned state dimension.

9.3 Response and adjoint evaluation package

The derivative study uses the predeclared dates, 12 objective–horizon cases, direction families, regional masks, physical inner products, and perturbation amplitudes. All models see the same states and perturbations. Three final seeds are used for the paired A/B/C/D comparison if affordable; seed spread is shown rather than folded into the test count. For each case save both physical-unit and normalized arrays, including state and control perturbations, masks, objective weights, JVPs, gradients, Taylor sequences, and line-search outcomes. Required displays and tables are:

1. **Perturbation validity:** central-difference linearity versus perturbation amplitude, plus/minus symmetry, numerical noise floor, and Taylor remainder slopes, separated by variable, depth, direction family, boundary/interior region, and horizon.
2. **Forward responses:** true, emulator, and difference response maps on common symmetric colour scales; predicted-versus-true directional-response scatter; relative response error, cosine, and gain ratio. Show every case as well as bootstrap intervals for the paired aggregate; do not hide sign failures behind the median.

3. **Adjoint fields:** signed MITGCM, emulator, and difference maps; depth slices and vertical profiles; full physical-space and declared projected-subspace relative error, cosine angle, magnitude ratio, sign agreement, and top-sensitivity overlap. Plot error versus wall distance and depth, and compare resolved gradient spectra/coherence to distinguish phase, amplitude, and boundary errors.
4. **Operator consistency:** verify the duality identity $q^T(Jv) = v^T(J^T q)$ for the emulator, compare automatic differentiation with finite differences, and report how ten-day Jacobian errors compose at 30 days.
5. **Control usefulness:** for each objective plot the predeclared line-search step, emulator-predicted ΔJ , forward-MITGCM ΔJ , and constraint residual. Store the full objective-versus-step curve, not only the chosen update.

Predeclared derivative targets, not promises, are:

- **Response:** D lowers blind response error by at least 25% versus its forward parent C while ten-day state error rises by no more than 5%.
- **Adjoint:** D improves median projected-gradient cosine by at least 0.2 versus C. Useful targets are median cosine above 0.8 at ten days and 0.6 at 30 days.
- **Usefulness:** at least 80% of predeclared emulator-gradient control steps lower the objective in forward MITGCM.

Failure remains diagnostic. Better one-step JVPs but poor 30-day adjoints implicate Jacobian composition; no JVP improvement implicates perturbation scale or subspace; interior-only success implicates boundary representation; good angles but wrong gains implicate calibration rather than sensitivity direction.

10 Ten-week execution plan

Table 9: Long MITGCM runs begin immediately and continue alongside code development.

Week	Activity	Output	Decision gate
1	Reproduce tutorial; benchmark; launch S0	Versioned setup, restart test, wall-clock/model-year	Runs are reproducible
2	Complete S0–S2; preprocess state maps	Common data store, masks, transpose-checked operators	Training trajectories pass QC
3	Audit and freeze A0; train ten-day baseline	Architecture manifest, small-sample overfit, A0 checkpoint	Adapted Bire map learns signal
4	Train modern Model A; spectrum study	A0/A forward comparison and baselines	Record controlled baseline
5	Train and freeze B	Rollout, spectrum, and boundary-loss ablation	Diagnose remaining forward failures
6	Select and freeze C	Learning curves, limited search, three seeds, complete forward package	Forward gate must pass
7	Run I1/I2; perturbation pilot and response data	Interpolation and linearity plots; training/validation responses	Perturbations resolved and local

Week	Activity	Output	Decision gate
8	Train D; freeze derivative comparison	C/D response ablation; sealed checkpoints and scripts	State skill preserved within 5%
9	Generate/verify adjoints; gradient and control tests	Taylor checks, maps, angles, and true objective changes	Open only after all freezes
10	Robustness, report, and handoff	Reproducible repository and manifest	Complete minimum claim

11 Risks and fallbacks

Table 10: Safeguards preserve a useful ten-week result.

Risk	Early diagnostic	Fallback
Spin-up cost or deep drift	Trends after year 50; measured cost	Scope first claim to upper ocean or use a documented equilibrated restart
Bire reconstruction ambiguity	Architecture audit exposes paper/repository conflicts	Freeze and label A0 as adapted; time-box historical repair
46-channel memory pressure	100-sample overfit and GPU profile	Reduce width, then TFNO, then vertical representation
FFT wraps across walls	Seam and boundary-response errors	More padding; Fourier continuation; local convolution and wall distance
Perturbation noise/nonlinearity	Central-difference pilot	Change ε by family before freezing it
Good ten-day, poor 30-day gradient	Horizon-separated metrics	Strengthen three-step training; report validated horizon honestly
Adjoint build delayed	Week-1 compilation smoke test	Deliver complete forward-response result; adjoint remains follow-on
Sampled directions too narrow	Good projected but poor full-map score	Add directions from response residuals as a labeled extension

The 0.25° configuration is a stretch goal only after C passes at 1° . A valuable extension is resolution transfer at 0.5° or 0.25° , testing both values and local responses. It must not displace the mandatory adjoint comparison.

12 Deliverables and reproducibility

The internship should leave:

1. version-controlled MITGCM configurations for S0–S2, I1–I2, perturbed restarts, and the 12 adjoint cases;
2. a manifest with checksums, units, grids, masks, split dates, perturbation vectors, and configuration hashes;
3. a frozen A0 architecture manifest and pinned NeuralOperator 2.0 pipelines for A0 and Models A–D;
4. separate sealed archives for forward training data and comparison-only adjoints;

5. scripts for forward, response, adjoint, and control-update metrics;
6. a report stating the validated state, subspace, horizons, objectives, and failures.

Record seeds, package lockfile, model checksum, normalization statistics, optimizer state, splits, and timing. Unit tests should cover staggered-grid interpolation and its transpose, gradient denormalization, masking, response construction, and a toy autodiff/adjoint identity.

13 Final scope statement

This project is feasible because it makes both the architecture-transfer and adjoint questions small and testable. It does not rebuild every undocumented detail of Bire et al.; A0 preserves the recoverable architecture and labels every forced adaptation, then the A–D ladder separates controlled modernization, forward selection, and derivative supervision on a documented MITGCM case. The project spends new simulation budget on 224 short, information-rich perturbations instead of more long trajectories, then reserves 12 expensive adjoints for a blind comparison.

The contribution is an empirical test of a transferable principle: *local forward responses can constrain a neural operator’s derivatives enough to improve adjoints for unseen objectives, without adjoint supervision.* The baseline, ablations, sealed evaluation, and explicit subspace claim make the outcome useful whether the hypothesis succeeds or fails.

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